

Diabetes Prediction using Machine Learning

Part 0 Introduction

1.

Diabetes is one of the most prevalent chronic diseases worldwide, presenting a critical public health challenge that demands data-driven approaches for early risk identification. This report focuses on developing and evaluating classification models to predict diabetes status using a cleaned dataset of 100,000 survey responses provided by Mohammed Mustafa, which consists of 8 potential predictors.

The plan to follow starts with exploring the data to understand patterns, followed by cleaning and preparing it. The core step is to fit multiple classification models—including logistic regression, decision trees and knn - and assess their goodness-of-fit using metrics like accuracy, precision, type 2 error and AUC-ROC to identify the optimal predictive classifier. After training and testing the models, the best one is chosen to support diabetes screening and medical decisions.

Part 1 Exploring variables and association

2.

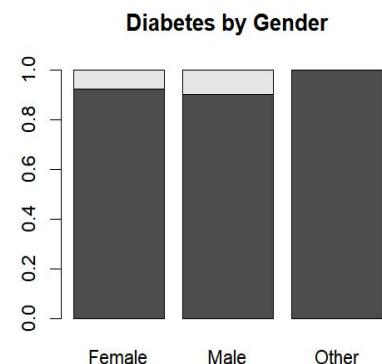
The binary response variable diabetes indicates whether a subject has been diagnosed with diabetes. I used prop.table and there are 91.5% of the sample classified as non-diabetic (0) and 8.5% as diabetic (1), showing a strong class imbalance with non-diabetic individuals forming the majority. This imbalance may affect model performance, so evaluation metrics beyond simple accuracy (such as sensitivity or recall) should be considered.

Among the input variables, demographic features include continuous age—a key risk factor for diabetes—and categorical gender (Female, Male, Other), while clinical binary variables hypertension and heart_disease capture health condition linked to diabetes risk. There are also variables showing subjects' habits that acts as potential predictors like categorical smoking_history variable, which can be simplified into 4 types (high, moderate, none, no info). Additionally, continuous numerical variables like HbA1c_level and blood_glucose_level serve as strong clinical predictors of diabetes status, collectively forming a comprehensive set of features for predictive modeling.

3.

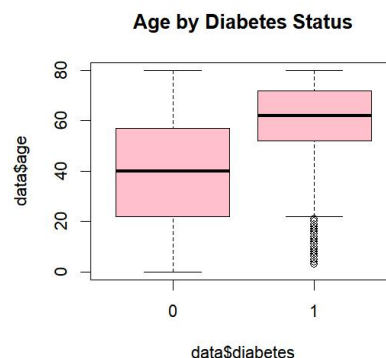
1) diabetes vs gender

The proportion of diabetes varies slightly across gender groups. Among female participants, 92.38% did not have diabetes and 7.62% had diabetes. Among male participants, 9.75% had diabetes. All participants in the Other gender category did not have diabetes. Overall, males show a slightly higher diabetes prevalence than females, while no cases of diabetes were observed in the Other group because of very few sample numbers. The relation is weak.



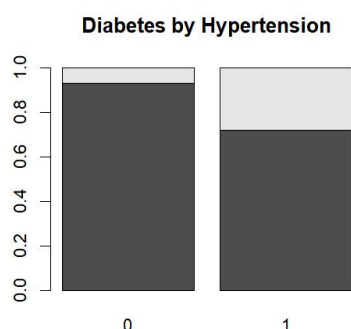
2) diabetes vs age

The box plot reveals a clear association between age and diabetes status. Diabetic participants exhibit a higher median age and an overall older age distribution compared to non-diabetic participants, whose ages are more concentrated in younger ranges. Diabetic participants' average age is 60.9, remaining skewed toward older ages, while this number for non-diabetes subjects is 40.1, indicating that older individuals are more likely to have diabetes in this dataset.



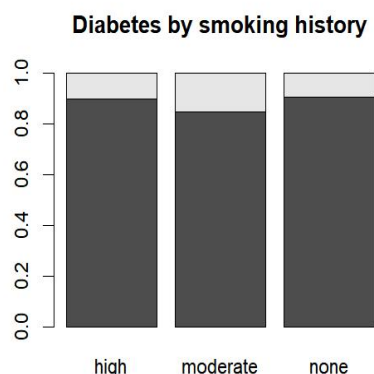
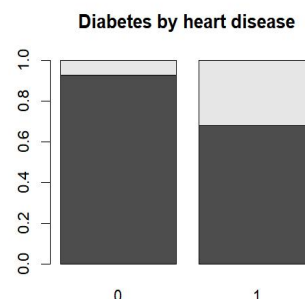
3) diabetes vs hypertension

The stacked bar plot demonstrates a strong positive association between hypertension and diabetes status. Among participants without hypertension, diabetic cases account for only 7%, while the proportion of diabetic cases is obviously larger in participants with hypertension, which is 28%. This suggests that individuals with hypertension are far more likely to have diabetes, highlighting hypertension as a significant predictor for diabetes risk.



4) diabetes vs heart_disease

The stacked bar plot reveals a strong positive association between heart disease and diabetes status. Among participants without heart disease, diabetic cases account for 7.5%, while the proportion of diabetic cases account for 32.14% among those with heart disease, with nearly one-third of this group being diabetic. This suggests that individuals with heart disease are at a much higher risk of having diabetes, marking heart disease as a very significant predictor for diabetes.

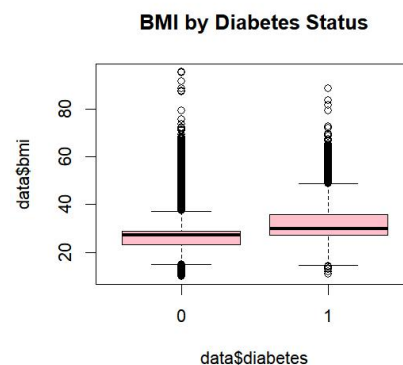


5) diabetes vs smoking_history

To simplify the data, the variable was into 4 parts to indicate intense smoking, moderate smoking and never smoking before respectively (no info group was filtered out). As shown in the figure, participants who never smoked before have a generally lower proportion of diabetes, only 9.7% compared to 15.4% for moderate smoking individuals. However, participants in intense smoking group seem to have lower proportion of diabetes (10.2%), which may be because of there's only 14.4% of samples are in this group, which leads to bias. In a word, smoking history is a very significant predictor for diabetes.

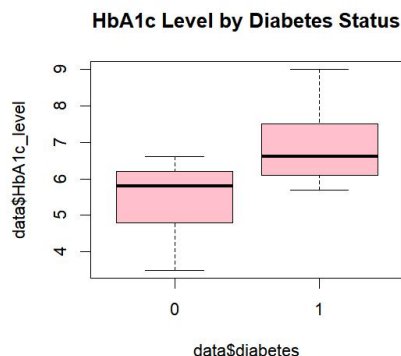
6) diabetes vs bmi

The box plot demonstrates a clear positive association between BMI and diabetes status. Diabetic participants exhibit a higher median BMI and an overall higher BMI distribution compared to non-diabetic participants, whose BMI are concentrated in lower ranges. While both groups include extreme high BMI outliers, the diabetic group's mean BMI is 32.0 compared to 26.9 in another group, indicating that higher BMI is strongly associated with diabetes.



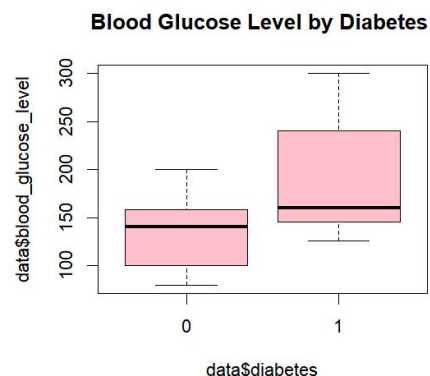
7) diabetes vs HbA1c_level

The box plot shows a very strong positive association between HbA1c level and diabetes status. Diabetic participants (mean = 6.9) exhibit higher median HbA1c level, with their entire distribution shifted upward relative to non-diabetic participants (mean = 5.4), whose HbA1c levels are concentrated in a lower range. The clear separation between the two groups highlights HbA1c level as a very significant predictor.



8) diabetes vs blood_glucose_level

According to common sense, blood glucose level is almost the decisive factor of diabetes. The box plot demonstrates a very strong positive association between blood glucose level and diabetes status. Diabetic participants, with the mean of 194.1, exhibit a substantially higher median blood glucose level. Their entire distribution shifted upward relative to non-diabetic participants, whose glucose levels are concentrated in a lower range with the mean of 132.9. The distinct separation between the two groups highlights blood glucose level as a highly predictive predictor for identifying diabetes status.



Part 2 Methods: KNN, DT, GLM

4.

1) . Logistic Regression (put it at first to get the optimal threshold)

For this model, all features in the dataset were initially included. Some individual categories, such as "Other" in gender, have p-values larger than 0.05; however, most categories within each feature show small p-values. I used this version of M1 to generate the ROC curve, AUC value (0.962) and the TPR-FPR versus threshold curve. As illustrated in the figure below, the optimal threshold is not the default 0.5. Instead, a threshold of 0.2 significantly increases sensitivity (TPR), capturing nearly all at-risk patients, while keeping the false positive rate (FPR) at a level that remains manageable through follow-up testing.

Considering that there are only 8.5% of the samples represent diabetic patients, the default 0.5 threshold is too conservative and results in many false negatives. For highly imbalanced datasets, the decision threshold should be tuned to prioritize the minority class, particularly when false negatives carry high cost.(Charles Elkan, 2001). Lowering the threshold to 0.2 increases the model's sensitivity, ensuring that more true diabetic cases are correctly identified.

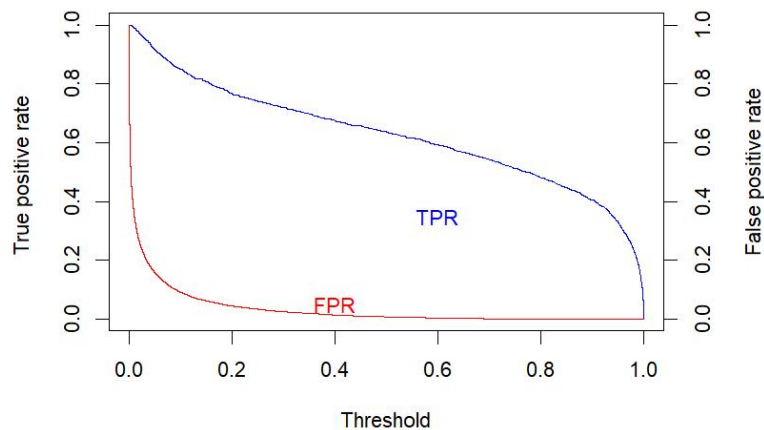


Figure 2.1 TPR–FPR versus threshold

After choosing the best threshold, 0.2 is used to calculate accuracy(0.939), tpr(0.766), fpr(0.045) of this model. Also, 0.2 will be used for both 3 models so as to unify the standards.

2) . Decision Tree

For Decision Tree, to get the best cp value, `cp_seq <- 10^seq(-6, -1, 1)` is used in the for loop. When the cp value decreases, the complexity of the tree grows with increasing tpr and decreasing fpr, tpr. I noticed that when cp is within 10^{-6} - 10^{-4} , acc and tpr fluctuate around 0.96 and 0.75, respectively, with fpr around 0.02. However, when $cp \geq 10^{-3}$, although the acc slightly rises to 0.97, the fpr drops to 0 straightly and tpr experienced a sharply decrease to 0.67. (As shown in the Figure 2,2)

This behavior indicates that the classifier is very simplistic and the model just separates almost all the samples as “no diabetes” roughly when the cp is too large. The slightly increase of acc and fpr = 0 correspond to the truth that the number of diabetes cases is very small, rather than true model performance. This result also implies that acc is misleading in this imbalance dataset. So we choose $cp = 0.0001$ as the best cp, which have high accuracy (0.964), high tpr (0.744) and an acceptable fpr (0.016).

Also, I'm curious about whether the threshold setting indeed improve the goodness of fit, so I also compute the acc, fpr and tpr under the default threshold of 0.5. (shown in blue in Figure 2.2). As shown in the figure, under the same cp value, tpr value of threshold of 0.5 is smaller compared to threshold of 0.2 group. Their acc don't have much difference unless the cp is too small. As for fpr, obviously, 0.5 threshold group performs slightly better because a threshold of 0.5 makes false positive predictions less likely. Overall, setting the threshold to 0.2 indeed help the model perform better.

The auc value is also calculated for this model, which is 0.964, indicating good model

performance.

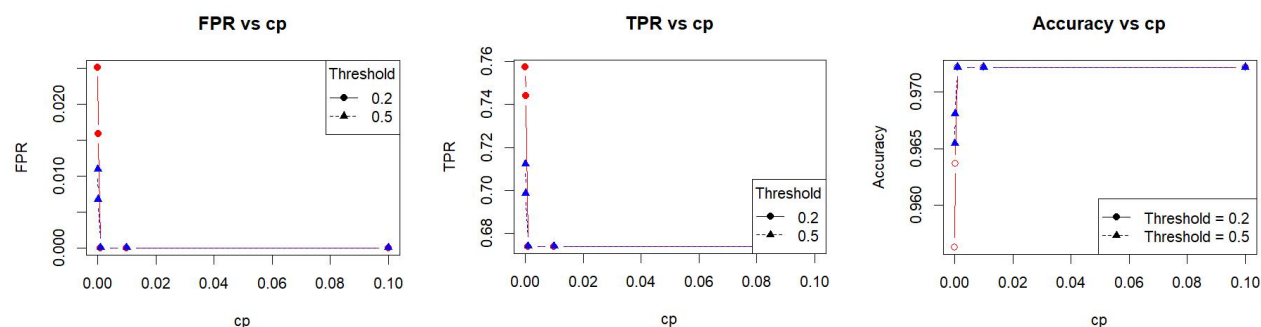


Figure 2.2 FPR/TPR/ACC vs cp (both for 0.2 and 0.5)

3) .KNN

For KNN, after preprocessing and scaling the dataset, I tested k values in $\text{seq}(1, 401, 2)$, where 401 was selected based on the $\sqrt{n}$ rule, which suggests that the optimal k is on the order of the square root of the sample size. After carefully evaluation, $k = 39$ stands out with the high tpr of 0.791 and strong acc of 0.94. Although some k values have slightly higher acc or lower fpr, considering the misleading effect of superficially good acc, fpr in such a severely imbalance dataset, $k = 39$ provides a better balance between sensitivity and overall performance, resulting in an AUC of 0.96. For comparison, a k value of 5, which demonstrates favorable accuracy and fpr, only got a auc of 0.908, indicating worse overall model performance.

The overall trend is similar to that observed in the Decision Tree: the 0.5 threshold group shows slightly higher accuracy, but the TPR drops to 0.56, meaning only 56% of diabetic patients are correctly identified—a level that is both risky and unacceptable. This result further confirms that selecting a threshold of 0.2 is a more appropriate choice for this model.

5.

To compare the 3 models, I list the GOF of best version for each model.

	LR	DT	KNN
tpr	0.766	0.744	0.791
auc	0.962	0.964	0.962
acc	0.939	0.964	0.943
fpr	0.045	0.016	0.043

Figure 2.3 Form for comparison

1). The best model

KNN achieves the highest true positive rate (0.791), indicating superior sensitivity in detecting diabetic patients, which is the primary concern in clinical screening, where missing a positive case could have serious health consequences. Beyond tpr, KNN maintains a high auc of 0.962, accuracy of 0.943 and a reasonable false positive rate of 0.043, demonstrating a well-balanced performance between correctly identifying diabetics and limiting

unnecessary alarms. Overall, being non-parametric, KNN captures complex, non-linear relationships that linear models like Logistic Regression may miss. Furthermore, KNN effectively handle the data without overfitting, unlike Decision Trees.

2) . Reason for selecting & Reference

A lot of materials and references have shown that prioritizing tpr is strongly supported in medical screening literature: Missing a diabetic patient (false negative) can lead to serious complications or delayed treatment. Screening programs are explicitly designed to maximize sensitivity, even if it increases false positives. World Health Organization screening principles emphasize that early detection systems should minimize missed cases. David L. Sackett (clinical epidemiology) highlights that sensitivity is critical when the cost of missing disease is high.

In evaluating diabetes prediction models, fpr is important but is treated as a controlled constraint rather than an independent optimization target. Accuracy, though intuitive, can be misleading in low-prevalence datasets such as diabetes, where only around 8.5% of patients are positive. A model that predicts all patients as non-diabetic could achieve over 90% accuracy while failing entirely to detect any true cases, demonstrating that accuracy does not reflect the clinical importance of false negatives.

Consequently, metrics like tpr and auc are preferred for evaluating models, as they better capture the balance between correctly identifying patients and overall discrimination performance (Elkan, 2001; Davis & Goadrich, 2006; Hand, 2009).

Part 3 Conclusion

6.

Based on the evaluation metrics, KNN with $k = 39$ is proposed as the best model for diabetes prediction due to its highest true positive rate and strong overall performance.

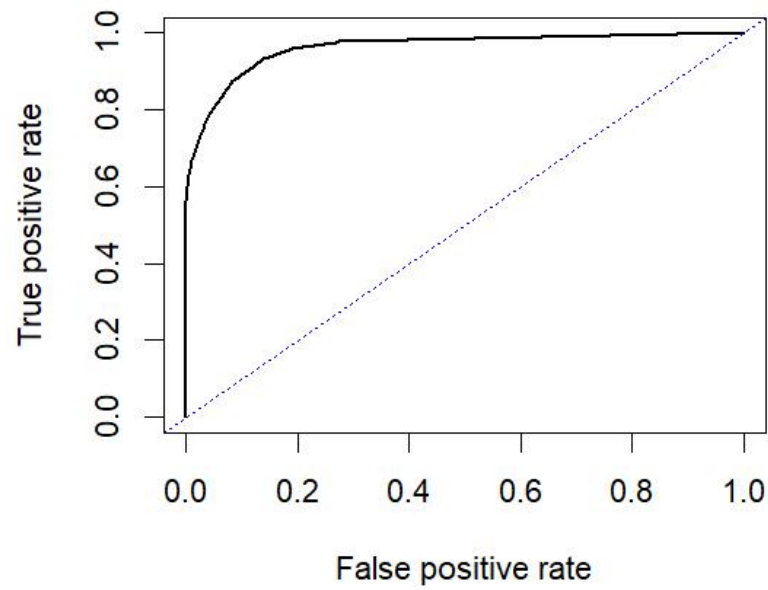
7.

1). LR Pros: Optimal for Threshold Tuning. Naturally produces probabilities, making it easy to adjust the classification threshold (e.g., 0.2 instead of 0.5) to prioritize sensitivity for diabetes screening, as we discussed earlier. Cons: Assumes Linear Relationships. Only consider the linear relationship between features and the response variable, which may miss complex, non-linear interactions.

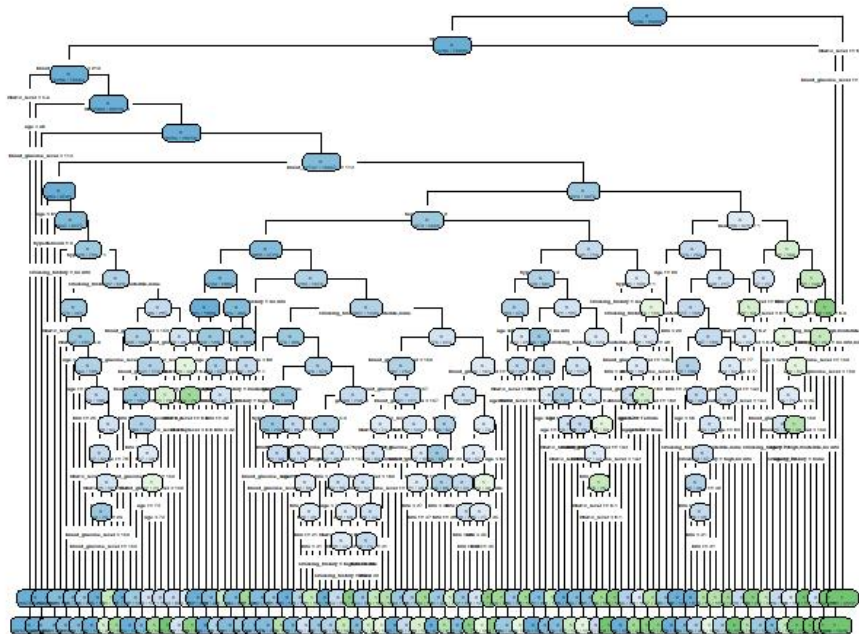
2). DT: Pros: Produces clear, human-readable decision rules. Naturally captures complex, non-linear associations and feature interactions without manual feature engineering. Cons: Prone to overfitting to noise in the training data, especially with deep trees, leading to poor generalization to new patient populations.

3). KNN: Pros: A non-parametric model that makes no assumptions about the underlying distribution of diabetes data, making it flexible for non-linear relationships. Cons: Slow to predict on large datasets, especially when trying different k , as it must compute distance to every training sample for each new patient. (took me 50 min to finish the loop)

APPENDIX



The ROC curve of the best model ($k = 39$)



The Design Tree ($cp = 0.0001$)
(Kind of overfitting)